

Book: Matrices, Schaum's Outline and related document, Lecture content

Ch-01: Defⁿ: Square Matrices, Equal Matrices, Zero Matrix; Practice: Sum of matrices, Multiplication Products by Partitioning; Equivalent of two matrices. Solved problems: 2, 4, 9(a);

Sup-problems: 11, 12, 13, 14, 17, 21

Ch-02: Defⁿ: Identify matrix, Diagonal matrix, Inverse of a matrix, Transpose of a matrix, Symmetric matrices, Skew-Symmetric matrix with examples, Conjugate of a matrix, Hermitian matrices with examples.

Solved-problems: 3, 5, 8, 9, 13,

Sup-problems: 21(a), 23, 24, 25, 26, 27(a)

32, 35, 36, 42(a), (b), 62

Ch-03: Find the determinant of a matrix, ex. 1

Solved problems: 1, 5, 8, 11;

Sup-problems: 17, 18, 19(a), (d), 27, 29, 37

Ch-05: Rank of a matrix, Elementary transformation of matrices, Equivalent matrices, Normal form of a matrix, elementary matrices, Row equivalence;

Solved Problems: 1, 3, 5, 6, 7

Sup-problems: 11, 15

→ Row and column transformation, converting Normal form and finding rank of a matrix

Ch-06: Finding adjoint A of a matrix,
 ex. 1, 2; solved problems: 2, 5, 4;

Sup-problems: 8, 11, 12, 14, 15, 20, 22

Ch-07: Calculate inverse of a matrix:

$A^{-1} = \frac{\text{adj } A}{|A|}$; ex. 1, 4, 3

solved problems: 2, 3, 5, 6

Sup-problems: 8, 11, 14, 20, 21, 22, 23

Ch-11: Vector spaces, How does a matrix represent a vector space? Determine that 2×2 matrices in a special form represent a vector space.

Instructions: Defⁿ: Square matrices, Equal matrix, Zero matrix, Identity matrix, diagonal matrix, Inverse of a matrix, Transpose of a matrix, Symmetric matrices, skew-symmetric matrix, Conjugate of a matrix, Hermitian matrix, Rank of a matrix, Elementary transformation of a matrix, Equivalent of matrices, Normal form of matrix, elementary matrix, Vector space.

Calculations: sum, multiplications, Hermitian, symmetric, skew-symmetric, matrix determinant, Inverse of matrix, transpose, Inverse by partitioning, Left inverse, right inverse, involutory matrix, Adjoint of matrix, Rank of matrix, transformation, Normal form & Vector space.

[proof of defⁿ of matrix relate]